

Some Practice Questions

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Optical Properties

Question 1

Compute the velocity of light in calcium fluoride (CaF_2), which has a dielectric constant ϵ_r of 2.056 (at frequencies within the visible range) and a magnetic susceptibility of -1.43×10^{-5} .

- Given that $\epsilon_r = 2.056$ and $\chi_m = -1.43 \times 10^{-5}$, we want to compute the velocity of light in calcium fluoride.
- We must first calculate the values of ϵ (permittivity) and μ (permeability) for calcium fluoride.

$$\epsilon = \epsilon_r \epsilon_0 = (2.056)(8.85 \times 10^{-12} \text{ F/m}) = 1.82 \times 10^{-11} \text{ F/m}$$

Now,

$$\begin{aligned}\mu &= \mu_0 (\chi_m + 1) \\ &= (1.257 \times 10^{-6} \text{ H/m})(1 - 1.43 \times 10^{-5}) = 1.257 \times 10^{-6} \text{ H/m}\end{aligned}$$

And, finally

$$\begin{aligned}v &= \frac{1}{\sqrt{\epsilon\mu}} \\ &= \frac{1}{\sqrt{(1.82 \times 10^{-11} \text{ F/m})(1.257 \times 10^{-6} \text{ H/m})}} \\ &= 2.09 \times 10^8 \text{ m/s}\end{aligned}$$

Question 2

The fraction of nonreflected radiation that is transmitted through a 10-mm thickness of a transparent material is 0.90. If the thickness is increased to 20 mm, what fraction of light will be transmitted?

Here, we are to calculate the fraction of non-reflected light transmitted through a 20 mm thick transparent material, given that the fraction transmitted through a 10 mm width is 0.90. Using the equation, $\ln \left[\frac{I'_T}{I'_o} \right] = -\beta x$; where the thickness is represented as x . The required fraction is just $\frac{I'_T}{I'_o} = e^{-\beta x}$.

$$\begin{aligned} \text{When } x = 10 \text{ mm,} \quad \beta &= -\frac{1}{x} \ln \left(\frac{I'_T}{I'_o} \right) \\ &= -\left(\frac{1}{10 \text{ mm}} \right) \ln(0.90) = 1.05 \times 10^{-2} \text{ mm}^{-1} \end{aligned}$$

Now, solving for $\frac{I'_T}{I'_o}$ when $x = 20 \text{ mm}$

$$\begin{aligned} \frac{I'_T}{I'_o} &= \exp(-\beta x) \\ \exp[-(1.05 \times 10^{-2} \text{ mm}^{-1})(20 \text{ mm})] &= 0.81 \end{aligned}$$

Magnetic Properties

Question 3

A coil of wire 0.20 m long and having 200 turns carries a current of 10 A.

- (a) What is the magnitude of the magnetic field strength H ?
- (b) Compute the flux density B if the coil is in a vacuum.
- (c) Compute the flux density inside a bar of titanium that is positioned within the coil.
- (d) Compute the magnitude of the magnetization M .

The susceptibility for titanium is 1.81×10^{-4} .

This problem concerns a coil of wire 0.20m long that has 200 turns and carries 10A.

(a) We calculate the magnetic field strength generated by this coil as:

$$\begin{aligned} H &= \frac{NI}{l} \\ &= \frac{(200 \text{ turns})(10 \text{ A})}{0.2 \text{ m}} = 10,000 \text{ A-turns/m} \end{aligned}$$

(b) In a vacuum, the flux density is determined as:

$$\begin{aligned} B_o &= \mu_o H \\ &= (1.257 \times 10^{-6} \text{ H/m})(10,000 \text{ A-turns/m}) = 1.257 \times 10^{-2} \text{ tesla} \end{aligned}$$

- c) When a bar of titanium is positioned within the coil, in order to compute the flux density given the magnetic susceptibility:

$$\begin{aligned} B &= \mu_o H + \mu_o M = \mu_o H + \mu_o \chi_m H = \mu_o H (1 + \chi_m) \\ &= (1.257 \times 10^{-6} \text{ H/m}) (10,000 \text{ A-turns/m}) (1 + 1.81 \times 10^{-4}) \\ &\cong 1.257 \times 10^{-2} \text{ tesla} \end{aligned}$$

- d) Magnetization is computed as follow as follows:

$$M = \chi_m H = (1.81 \times 10^{-4}) (10,000 \text{ A-turns/m}) = 1.81 \text{ A/m}$$

Question 4

The magnetic flux density within a bar of some material is 0.435 tesla at an H field of 3.44×10^5 A/m. Compute the following for this material:

- (a) the magnetic permeability, and
- (b) the magnetic susceptibility.

Here, the magnetic flux density (B) within a bar of some material is 0.435 tesla at a magnetic field strength (H) of $3.44 \times 10^5 \text{ A/m}$.

a) We calculate the magnetic permeability (μ) as:

$$\mu = \frac{B}{H} = \frac{0.435 \text{ tesla}}{3.44 \times 10^5 \text{ A/m}} = 1.2645 \times 10^{-6} \text{ H/m}$$

b) The magnetic susceptibility is then calculated as: $\chi_m = \mu_r - 1$

$$\begin{aligned}\chi_m &= \frac{\mu}{\mu_0} - 1 = \frac{1.2645 \times 10^{-6} \text{ H/m}}{1.257 \times 10^{-6} \text{ H/m}} - 1 \\ &= 6 \times 10^{-3}\end{aligned}$$

Electrical Properties

Question 5

- a. Calculate the resistance of an aluminum wire 5 mm in diameter and 5 m in length. The conductivity of aluminum is $3.8 \times 10^7 (\Omega\text{-m})^{-1}$
- b. What would be the current flow if the potential drop across the ends of the wire is 0.04 V?
- c. What is the current density?
- d. What is the magnitude of the electric field across the ends of the wire?

We solve for the resistance in terms of the conductivity, length and diameter as follows:

$$\begin{aligned} R &= \frac{\rho l}{A} \\ &= \frac{\left(\frac{1}{\sigma}\right)l}{A} = \frac{l}{\sigma A} \\ &= \frac{l}{\sigma \pi \left(\frac{d}{2}\right)^2} \end{aligned}$$

Applying values of conductivity ($3.8 \times 10^7 (\Omega\text{-m})^{-1}$), diameter (5 mm) and length (5 m), we get:

$$\begin{aligned} R &= \frac{l}{\sigma \pi \left(\frac{d}{2}\right)^2} \\ &= \frac{5 \text{ m}}{\left[3.8 \times 10^7 (\Omega\text{-m})^{-1}\right] (\pi) \left(\frac{5 \times 10^{-3} \text{ m}}{2}\right)^2} \\ &= 6.70 \times 10^{-3} \Omega \end{aligned}$$

(b) If $V = 0.04 \text{ V}$ then, the current is computed as follows:

$$I = \frac{V}{R} = \frac{0.04 \text{ V}}{6.70 \times 10^{-3} \Omega} = 6.00 \text{ A}$$

(d) The electric field is computed as follows:

$$E = \frac{V}{l} = \frac{0.04 \text{ V}}{5 \text{ m}} = 8.0 \times 10^{-3} \text{ V/m}$$

(c) The current density is equal to the current divided by the specimen cross-sectional area as:

$$J = \frac{I}{A} = \frac{I}{\pi \left(\frac{d}{2} \right)^2} = \frac{6.00 \text{ A}}{\pi \left(\frac{5 \times 10^{-3} \text{ m}}{2} \right)^2} = 3.06 \times 10^5 \text{ A/m}^2$$

Question 6

At room temperature the electrical conductivity and the electron mobility for aluminum are $3.8 \times 10^7 (\Omega \cdot \text{m})^{-1}$ and $0.0012 \text{ m}^2/\text{V} \cdot \text{s}$, respectively.

Calculate the number of free electrons per cubic meter for aluminum at room temperature.

The number of free electrons per cubic meter for aluminum at room temperature may be calculated as follows:

$$\begin{aligned} n &= \frac{\sigma}{|e| \mu_e} \\ &= \frac{3.8 \times 10^7 (\Omega\text{-m})^{-1}}{(1.602 \times 10^{-19} \text{ C})(0.0012 \text{ m}^2/\text{V-s})} \\ &= 1.98 \times 10^{29} \text{ m}^{-3} \end{aligned}$$

Question 7

At room temperature the electrical conductivity of PbS is $25 (\Omega \cdot \text{m})^{-1}$, whereas the electron and hole mobilities are 0.06 and $0.02 \text{ m}^2/\text{V} \cdot \text{s}$, respectively. Calculate the intrinsic carrier concentration for PbS at room temperature.

In this problem we are asked to calculate the intrinsic carrier concentration for PbS at room temperature.

Since the conductivity of both electron and hole mobilities are provided in the problem statement, all we need do is solve for n or p (i.e., n_i which is equal for intrinsic semiconductors) as follows:

$$\begin{aligned} n_i &= \frac{\sigma}{|e|(\mu_e + \mu_h)} \\ &= \frac{25 (\Omega\text{-m})^{-1}}{(1.602 \times 10^{-19} \text{ C})(0.06 + 0.02) \text{ m}^2/\text{V-s}} \\ &= 1.95 \times 10^{21} \text{ m}^{-3} \end{aligned}$$